Honours Report Spring ‘24

PastPaths: Mapping Historical Trade from Texts

:- Hitesh Goel
2020115003
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Introduction

In this report, I will be writing out the details of the project I have done which has now been accepted at the Turkey Computational Social Science Conference 2024.

We develop a pipeline to analyse historical trade networks from textual data. Given a text document, we are able to generate a cartographic map showcasing the trade of commodities at various places mentioned in the document. Our idea is to provide historians with valuable insights to enhance their qualitative analysis using state-of-the-art computational tools. A lot of historical research is done analysing trade networks and even the movement of people, groups, etc throughout history. However, since all of this analysis is strictly qualitative, a researcher can only look at 5-10 documents during their research. While this allows them to do detailed studies on the various patterns observed in the given documents, they cannot always back it up using quantitative methods. Our project aims to bridge this gap between the quantitative and the qualitative methods. Using the pipeline we present, the researchers will be able to look at and visualise trade networks from a large corpus of data with relative ease. This is a first of its kind study in the intersection of the Social Sciences and Computer Science, where we attempt to bridge the gap between the two fields and showcase how CS techniques can be used to better answer research questions grounded in Social Science.We propose a method capable of identifying and mapping the trade of commodities across the Western Himalayas. Additionally, our approach enables a temporal comparison of trade evolution throughout the 19th and 20th centuries. Now I shall present how we went about building the pipeline.

Overview

Let us first take a high-level look at the work we have done and then we shall go into the details of each of the steps.

The place names, people, and the commodities in the text are identified using Named Entty Recognition (NER), which is a standard method in NLP for such classification tasks.

To be able to perform NER, we first had to create our own dataset for the task. Given that the domain of our study is fairly niche, with us only focussing on historical texts centred around the Western Himalayas, there is a dearth of such freely available datasets. Not only this, our specific task involves classifying the place names, commodities as well as the people that occur in a text. While there are quite a few datasets that are built for the classification of people and places, as to our knowledge, none of them have any commodities data available. Hence, annotating a new dataset from scratch was a necessity. We'll talk more about the details of this dataset in the next slides.

Then the next step was to train custom NER models using Deep Learning techniques. A SpaCY model is used as the baseline, while we train multiple other models for our classification task.

Once we train the NER model, we can now get the place names and the commodities that occur in a piece of text. The next step is to find out which commodities are traded by which places. This is modelled as a Relation Extraction model in NLP. Once again, we did not have any datasets available to train a custom Relation Extraction model. This time, instead of annotating a new dataset from scratch, we decided to leverage the capabilities of Large Language Models (LLMs) like GPT-3.5 and Llama-2. We pass the piece of text, all of the places and commodities that occur in the text to the LLM and prompt it to return the required relations i.e. which places trade which commodities. Several prompting strategies like few shot prompting (where we provide some gold standard outputs along with the prompt) and Chain Of Thought prompting were also tried out.

Finally, after obtaining the relations, we generate the cartographic maps. The coordinates of the places are extracted from the GeoNames gazetteer and plotted on the map using basic Python libraries.

Literature Review

Before diving into the details of each of the steps, we shall first look at some of the past work in the domain. Since this is a novel research problem, there is not a lot of research work or end-to-end studies for our particular use case. However, there is still a lot of literature available that help us in all of the separate components that we use in our pipeline.

The one main paper or research work in this domain is the Edinburgh Geoparser developed by the Language Technology Group at Edinburgh University, It is the most well known NER model for classifying places in historical text data. In their model, they have mainly used traditional NLP techniques and other lingusitic features and have not used Deep Learning strategies to train their NER. They also use multiple historical gazetteers available to them to make a list of the possible place names that can occur in the text and make use of this database in the model. We could not directly use this in our project because they do not classify commodities that occur in the text. Hence, we had to train our own NER model. Also, since we did not have access to all the detailed gazetteers as them, we decided to leverage Deep learning techniques to be able to account for out-of-vocabulary words, or in other words, to be able to classify places which our model may not have seen.

With the advent of transformers, the NER task using deep learning has seen a lot of progress in recent years and a lot of research has been published in the domain with the papers claiming extremely good performances on previously defined NER datasets. Most of these papers use small Language Models like BERT and RoBERTa which are transformer encoder models. Along with them, some research also suggests using a BiLSTM-CRF layer for better classification of text. Hence, these are the models that we train for our particular dataset.

Another important technique that we utilise is the domain adaptive pretraining method. This was proposed recently and is used in cases where wish to tune our language models to a particular domain or field. The idea is that LMs like Bert are trained on Wikipedia data and hence have never encountered original historical texts, so their performance on such data is going to be worse. To counter this, we once again pretrain our Bert model on a large corpus of historical data so that it starts to learn the linguistic features of that particular domain and results show that we do in fact achieve better results using this technique.

Dataset Characteristics

Our dataset, rooted in the 19th-century Western Himalayas, comprises of 22 historical documents meticulously annotated for the Named Entity Recognition (NER) task. We identify three primary named entities: Place Names, People, and Commodities, addressing the scarcity of specialised datasets in the domain of NER for historical texts. Annotations were done by two students over a period of three months using Prodigy. A Cohen's Kappa of 0.83 and an F1 score of 0.96 was achieved in the inter-annotator agreement.

Here, we have also provided a brief description of each of the entities and how they are to be identified while annotating the documents:

Entity	**Description**	**Examples**
Place Names	Geographical identifiers denoting specific locations within the Western Himalayas.	Garhwal, Kumaon,Simla
People	Individuals and groups mentioned within the document.	Sayid, Gujur, Kuchis
Commodities	Goods and products indicative of the economic activities and trade prevalent in the region.	Wheat, rice, shawl

Here are the final statistics of our dataset:

Documents	Sentences	Tokens	Place Names	People	Commodities
22	69419	653220	13650	5197	1149

Methodology

The initial step of our project involves converting digitized PDFs into text data using OCR methods. Standard OCR libraries such as PyPDF2 in Python often yield suboptimal results due to the patchy nature of text in these documents. Therefore, we utilize GoogleOCR, which is one of the best OCR models for this task as it employs advanced AI techniques to extract clean and accurate text from these documents.

Once the text is extracted, we proceed to train our custom Named Entity Recognition (NER) models. The annotations for these models were performed using Prodigy, a tool that provides a SpaCy baseline for NER tasks. Beyond this baseline, we train three additional models: a BiLSTM+CRF architecture, a BERT + BiLSTM + CRF architecture, and a domain-adapted BERT + BiLSTM + CRF architecture. The domain-adaptive pretraining of BERT is particularly important for improving performance on historical data. For this task, we selected a separate set of 50 historical documents from the 19th century and employed standard BERT training processes, including Masked Language Modeling and Next Sentence Prediction, to pretrain the model.

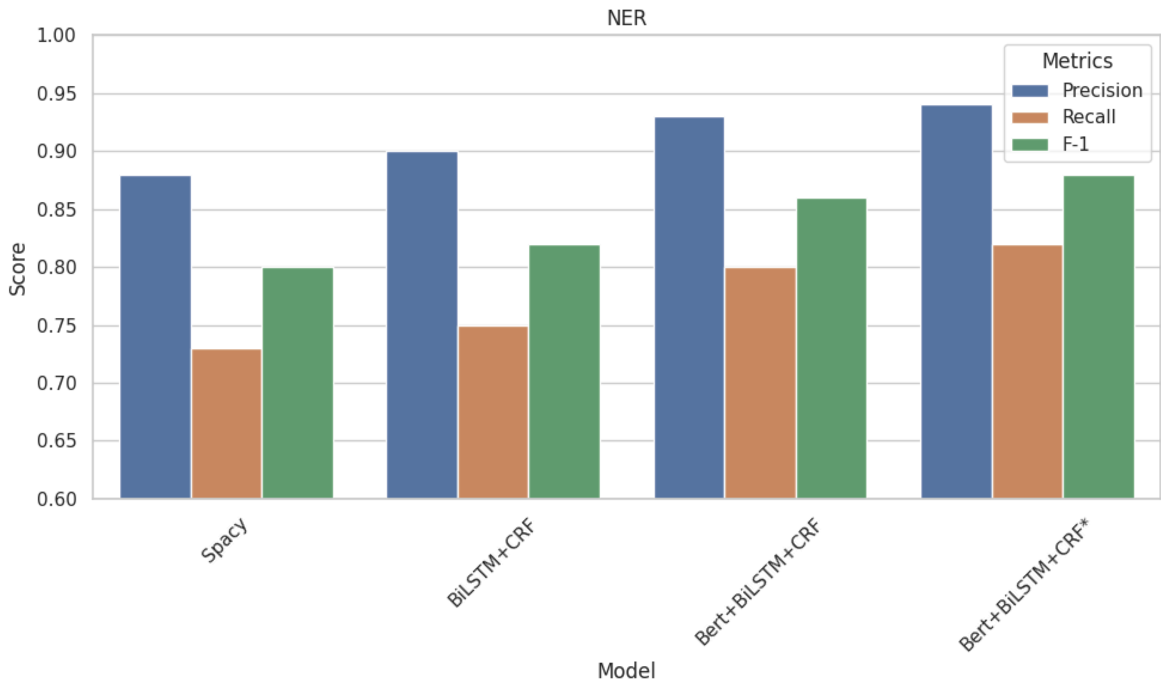
Given the small size of our dataset, we hypothesized that using very large language models like GPT would be impractical, as these models typically require substantial amounts of data to train effectively. Therefore, we focused on models better suited to our dataset size and specific requirements.

For relation extraction, we leverage the capabilities of large language models (LLMs) due to the lack of suitable relation extraction datasets. We employ diverse prompting strategies such as zero-shot prompting, few-shot prompting, and chain-of-thought prompting to enhance performance. We test multiple LLMs for this task, including GPT-3.5 via the OpenAI API, Llama-2-7b (an open-source model released by Meta last year), and Unlimiformer + Llama-2. The Unlimiformer model, introduced in a paper published last year, extends the context length from 4k to very long contexts (up to 500k). The major advantage of using long context models like Unlimiformer + Llama-2 is the ability to process larger pieces of text at once, providing the LLM with a more extensive context for relation extraction. We hypothesize that this approach will yield better performance compared to using the simple Llama-2 model.

Finally, to generate maps from the extracted relations, we use coordinates of places obtained from the GeoNames gazetteer. This step ensures that the spatial information is accurately represented, completing the process from text extraction to spatial mapping.

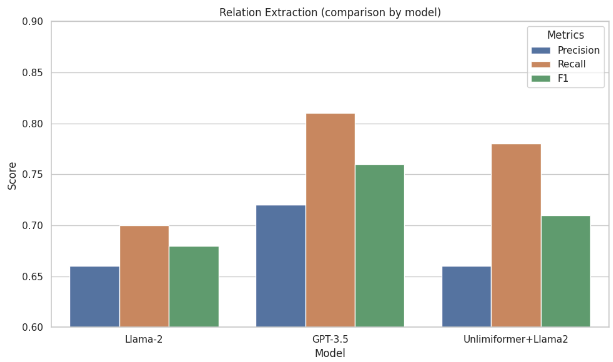
Results

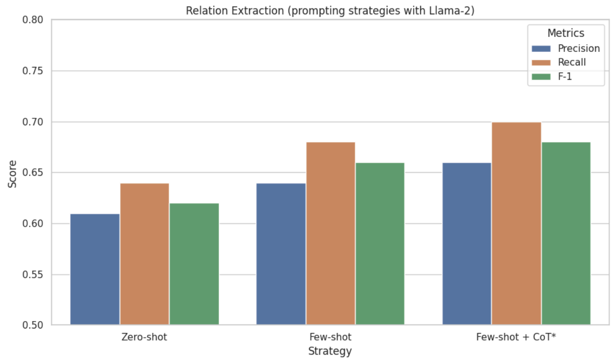
NER:



All of our custom trained models outperform the SpaCY baseline. We get the best results for the Bert-BiLSTM-CRF model with the adaptive pretrained Bert. This further proves the efficacy of pretraining bert on domain specific data so as to achieve better results.

RELATION EXTRACTION:





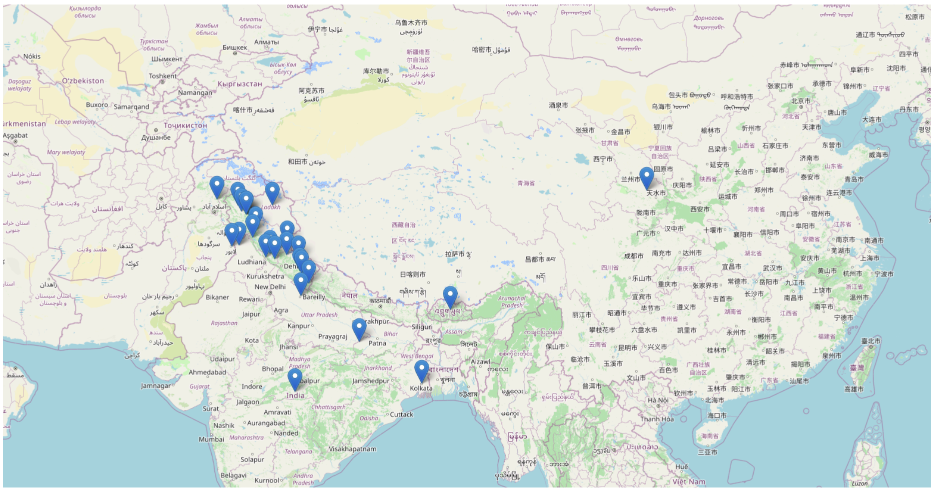
For performing this evaluation, we faced a challenge due to the lack of an existing dataset. To address this, we selected four random documents from our initial set of twenty-two and manually built all possible relationships within the text. This manually annotated set is considered our gold standard, against which we evaluate the performance of our LLMs. Although this evaluation strategy is not perfect, we believe that demonstrating good performance on this dataset indicates the effectiveness of using LLMs for the relation extraction task through prompting.

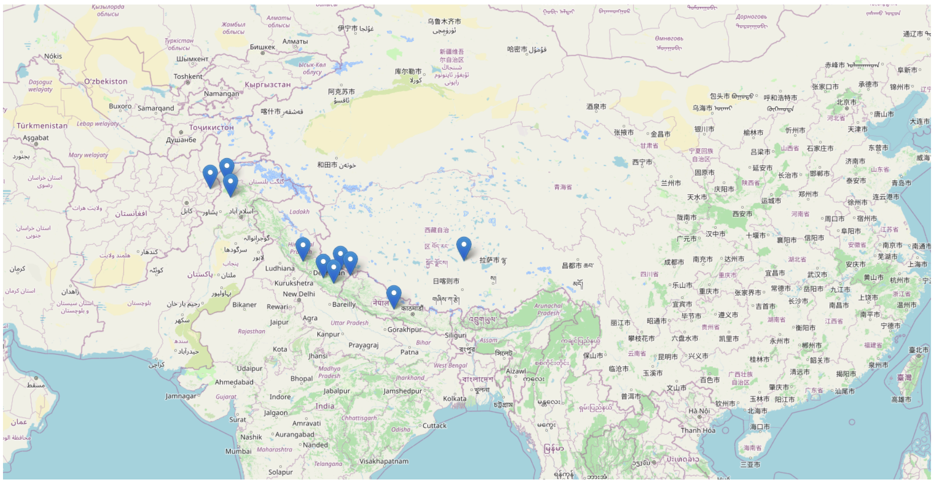
In our experiments, GPT-3.5 performed the best in the relation extraction task compared to all the other models tested. Notably, extending Llama-2 for long contexts significantly improved the model's recall. As hypothesized, the model's ability to process larger pieces of text at once allows it to capture more relationships, thus improving recall. This extension helps the model avoid missing certain relations that were previously overlooked when working with shorter text segments.

When comparing different prompting strategies, we found that the few-shot combined with chain-of-thought (CoT) method yielded the best results. This strategy enhanced the model's performance by providing it with a better framework to understand and extract the relationships from the text accurately. Overall, these findings highlight the importance of both model selection and prompting strategies in optimizing the performance of LLMs for relation extraction tasks.

Visualisations

These are the visualisations or the final maps that we obtain using our model. Over here, we have presented a sample output showcasing the trade of cloth in the Western Himalayas at two different time periods: the beginning of the 19th century (1810) and the end of the 19th century (1890s). The idea is that we pick documents talking about the trade in the region from the first 10 and the last 10 years of the century and pass them to our pipeline to generate these maps. These documents picked are different from the ones used in the training of the NER model. As we can see, we are able to build neat visualisations of the places that traded a particular commodity during a certain time period.





Future Work

Here are some advancements we are currently working on to enhance our project. We are focused on improving map detail by classifying places as importers or exporters and building trade networks where edges denote the direction of commodity flow. This enhancement can be modeled as a knowledge graph problem in computer science. To achieve this, we are fine-tuning our NER model to classify places accurately as importers or exporters. Additionally, we are constructing a knowledge graph (KG) dataset to train our model in identifying the direction of imports.

The current NER model has primarily been trained on a corpus of documents centered around the Western Himalayas. Our goal is to develop a more generalized model capable of classifying entities in any historical document. However, the lack of diverse datasets remains a significant bottleneck in this effort.

Furthermore, we aim to deploy an open-source application that will enable researchers to upload text documents and generate maps in real-time. This tool will facilitate broader access to our methodology and support historical research by providing detailed and accurate visual representations of trade networks and other spatial relationships.

Conclusion

We have explored a novel research problem at the intersection of history and computer science. Our proposed pipeline is capable of generating maps that represent historical trade networks from a given set of textual resources. The low running costs and scalability of our pipeline are particularly promising, as they enable us to deploy the project on a large scale, making it accessible to researchers worldwide. This approach not only advances the study of historical trade but also provides a valuable tool for historians and other scholars.

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